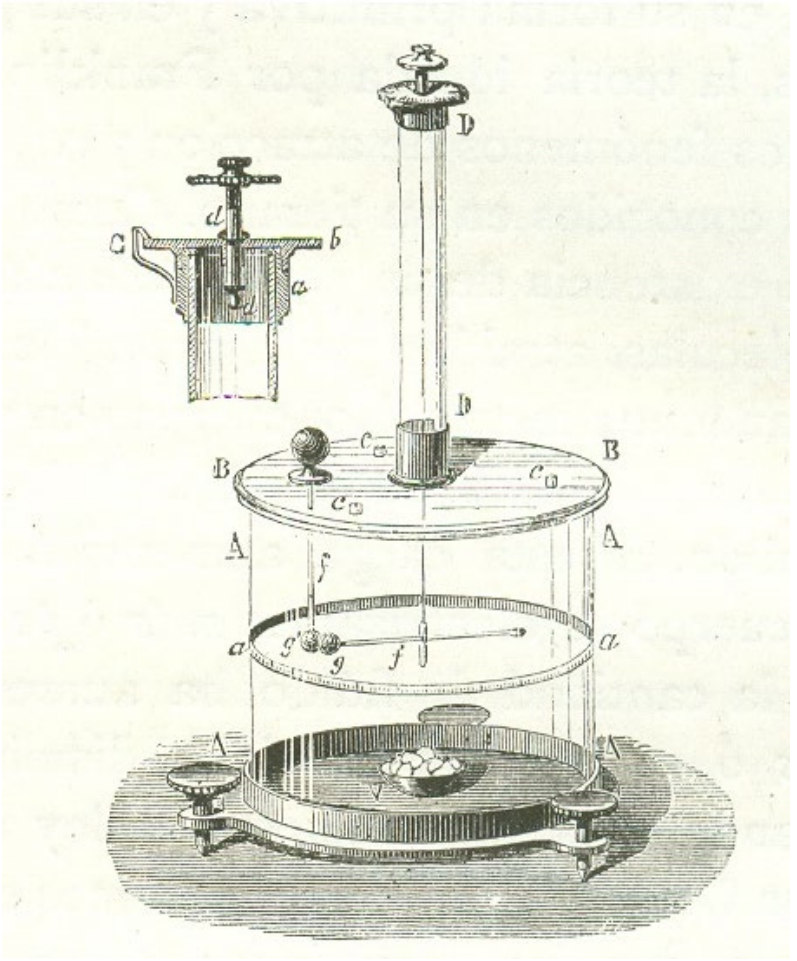
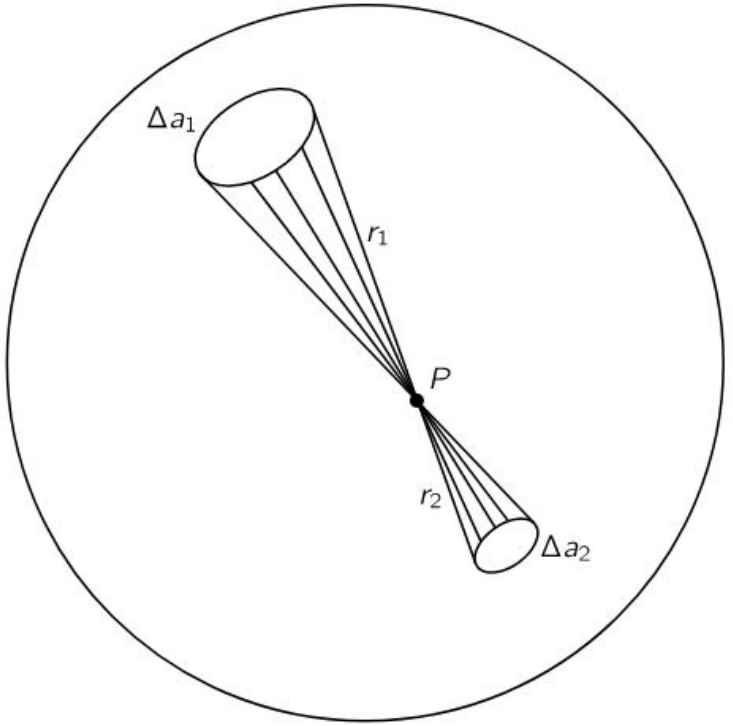


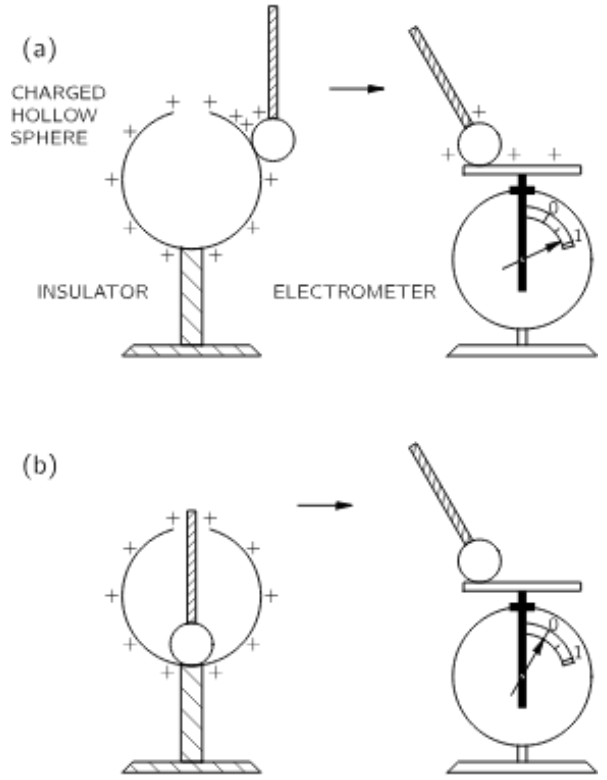
Charles-Augustin de Coulomb
(1736 – 1806)





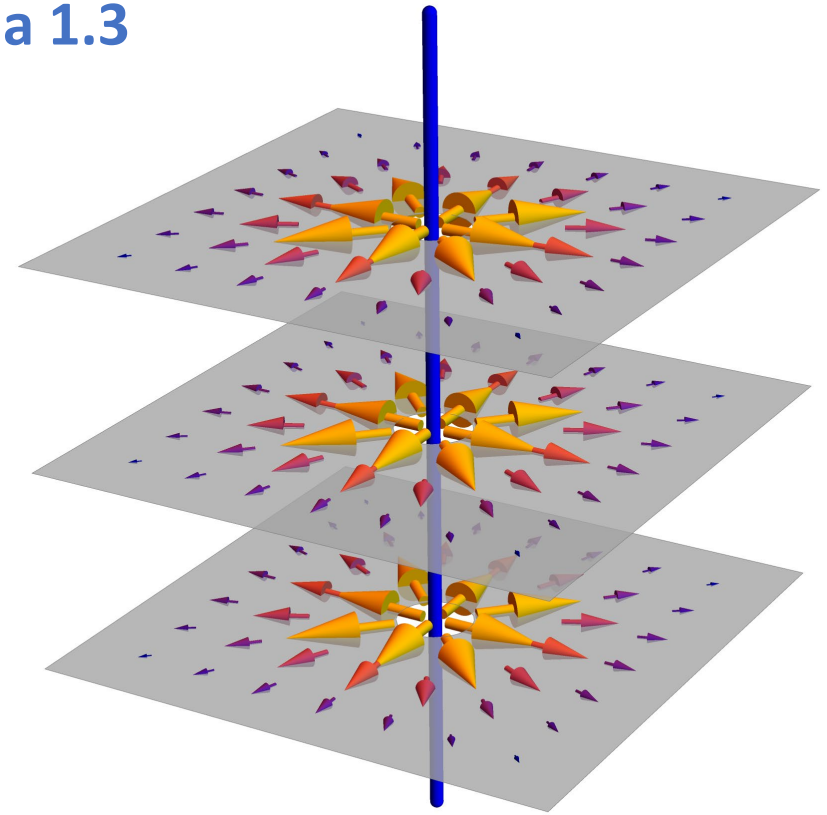
Henry Cavendish
(1731 - 1810)

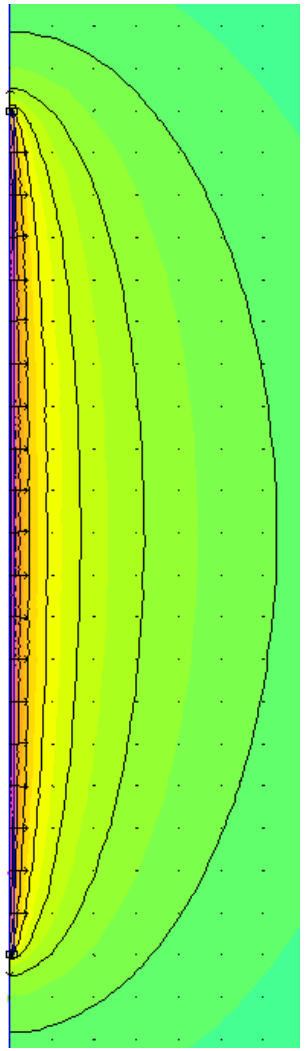




The Feynman Lectures on Physics, Vol. 2.
5.8. Is the field of a point charge exactly $1/r^2$?
http://www.feynmanlectures.caltech.edu/II_toc.html

Problema 1.3

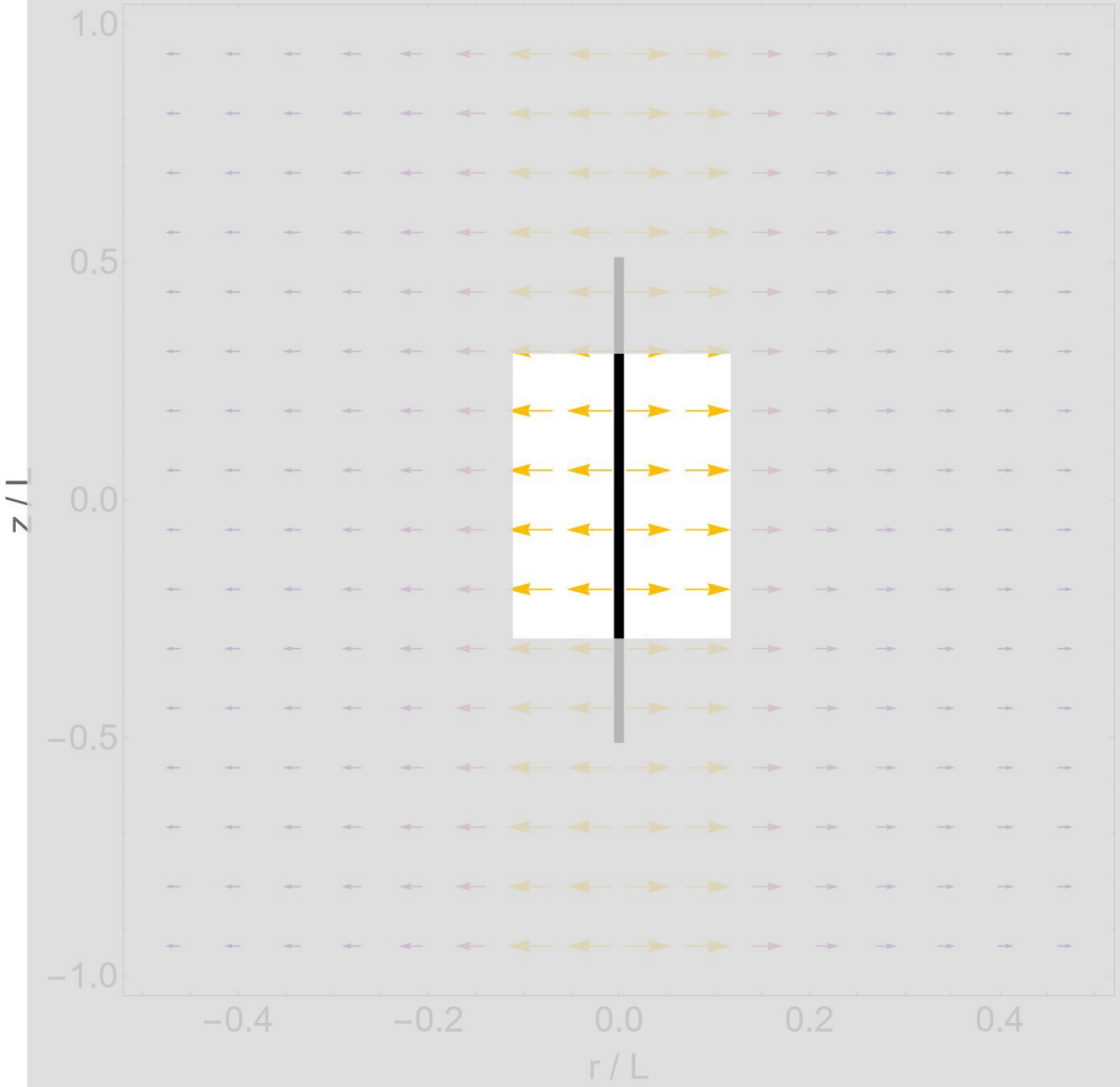


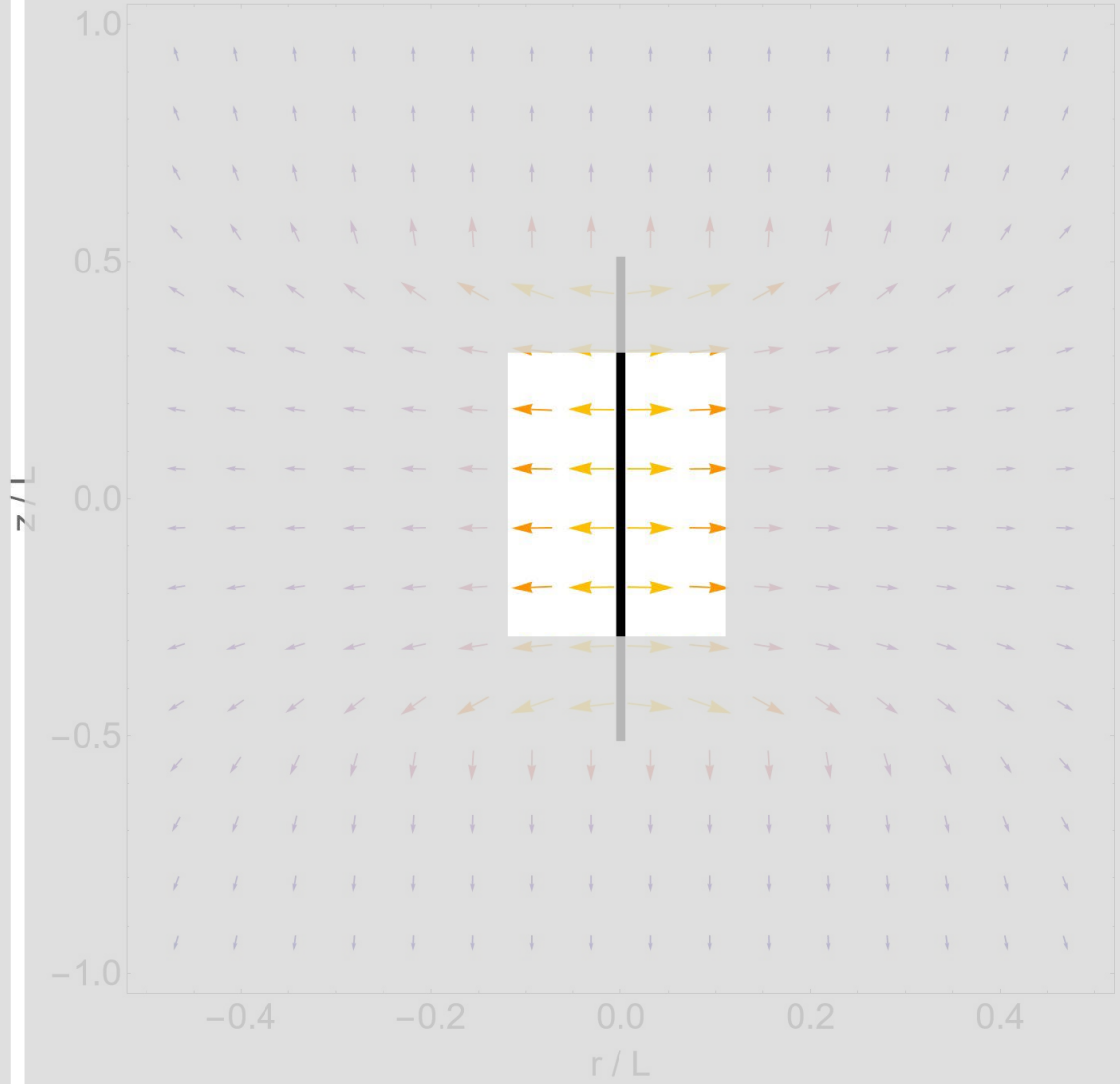


$$\vec{E}_{ap}\,(\vec{r}) = \frac{\lambda}{4\pi\varepsilon_0 r}\hat{u}_r$$

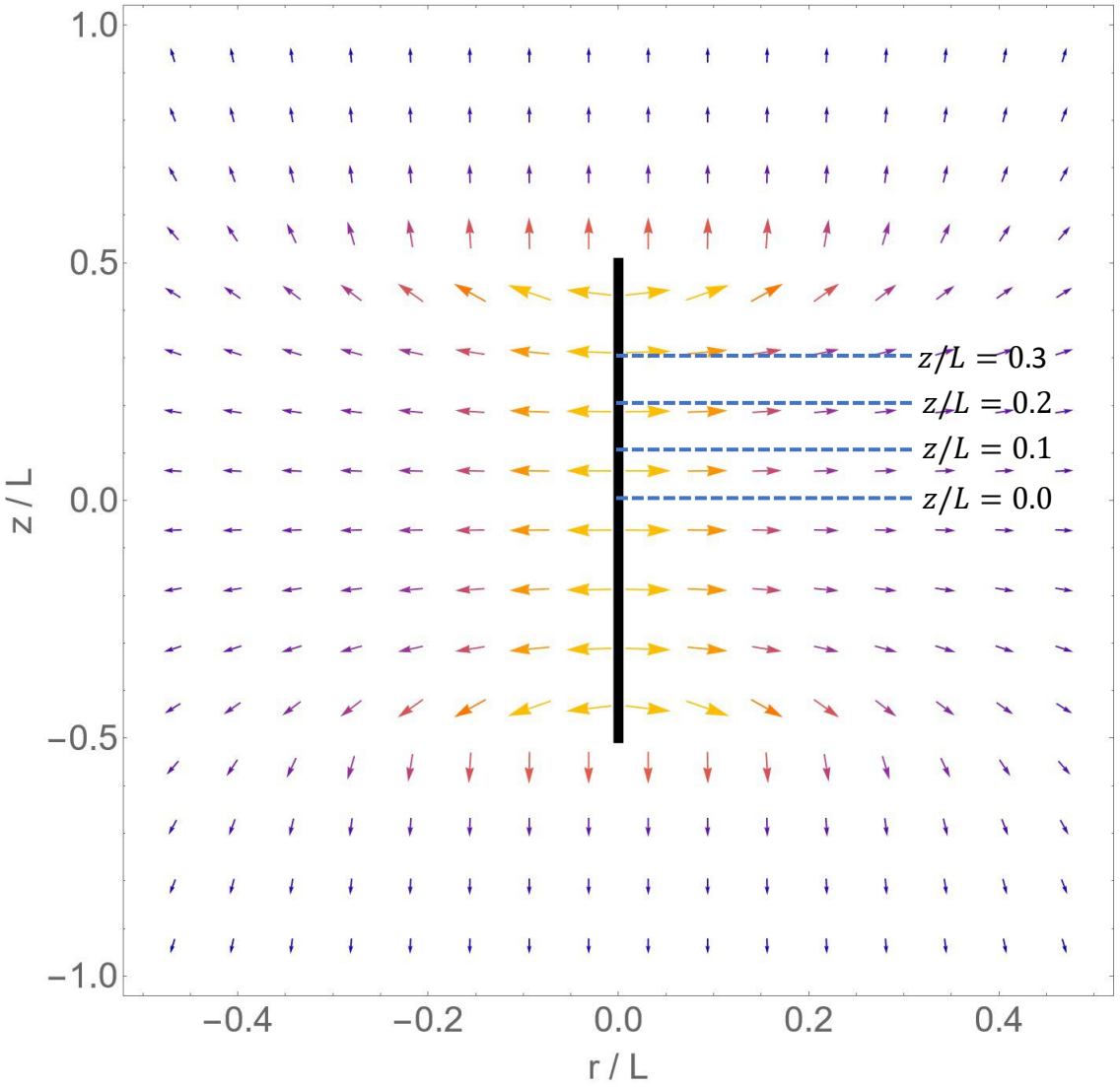
$$\vec{E}_{ex}(\vec{r}) = \frac{\lambda}{4\pi\varepsilon_0 r}\left(\frac{z+L/2}{\sqrt{r^2+(z+L/2)^2}} - \frac{z-L/2}{\sqrt{r^2+(z-L/2)^2}}\right)\hat{u}_r - \frac{\lambda}{4\pi\varepsilon_0}\left(\frac{1}{\sqrt{r^2+(z+L/2)^2}} - \frac{1}{\sqrt{r^2+(z-L/2)^2}}\right)\hat{u}_z$$

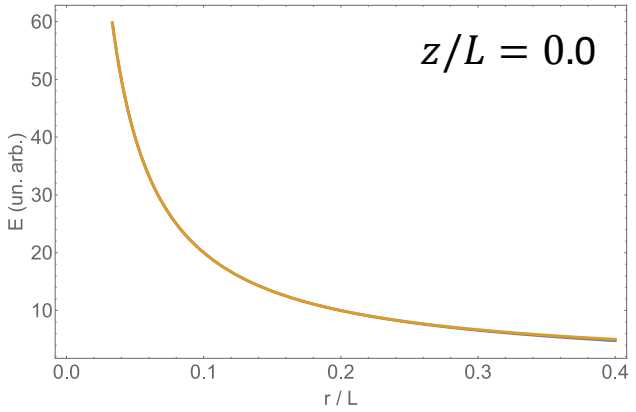
Problema 1.3

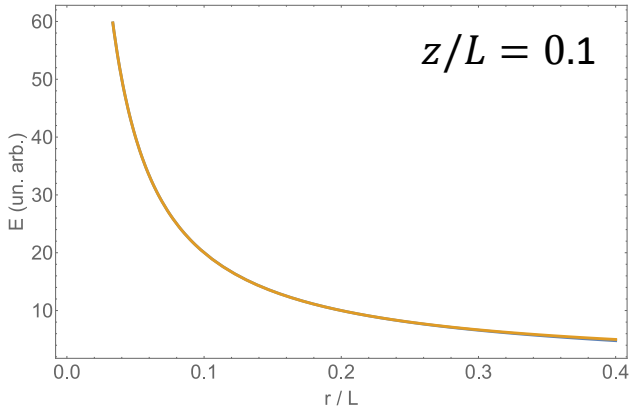


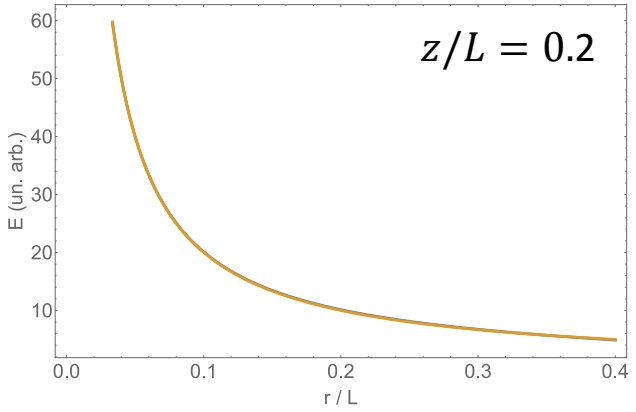


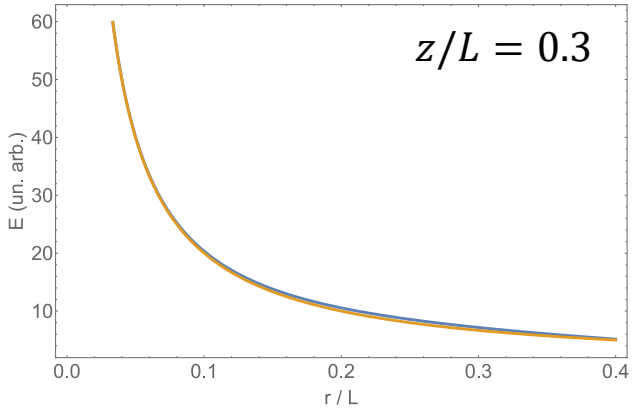
Problema 1.3


1.0
0.5
0.0
−0.5
−1.0
z / L
−0.4
−0.2
0.0
0.2
0.4
r / L
$z/L = 0.3$
$z/L = 0.2$
$z/L = 0.1$
$z/L = 0.0$

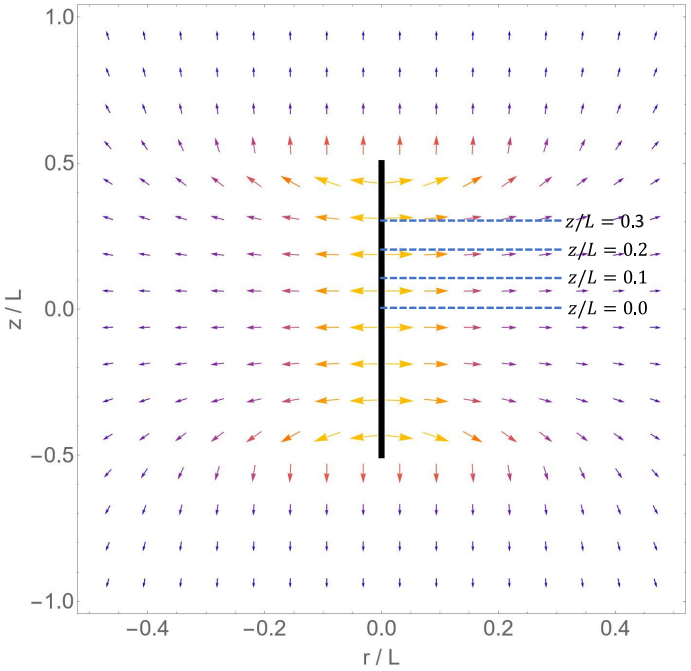

$z/L = 0.0$
E (un. arb.)
60
50
40
30
20
10
0.0
0.1
0.2
0.3
0.4
r / L

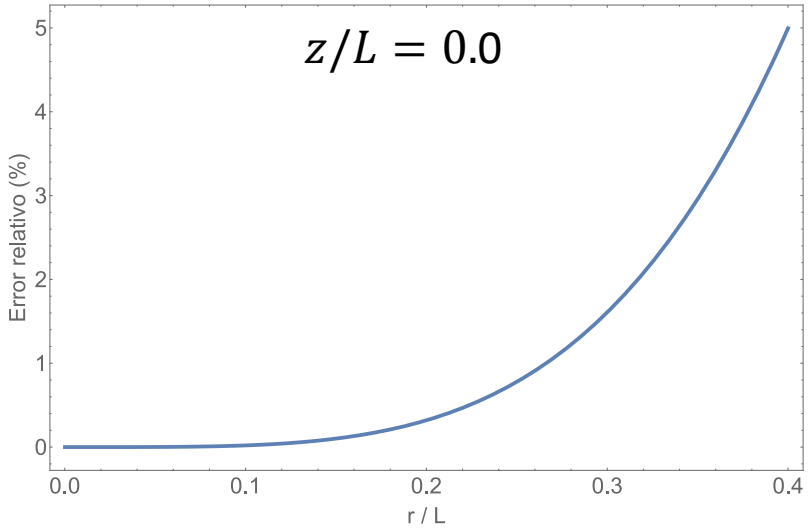

$z/L = 0.1$
E (un. arb.)
60
50
40
30
20
10
0.0
0.1
0.2
0.3
0.4
r / L

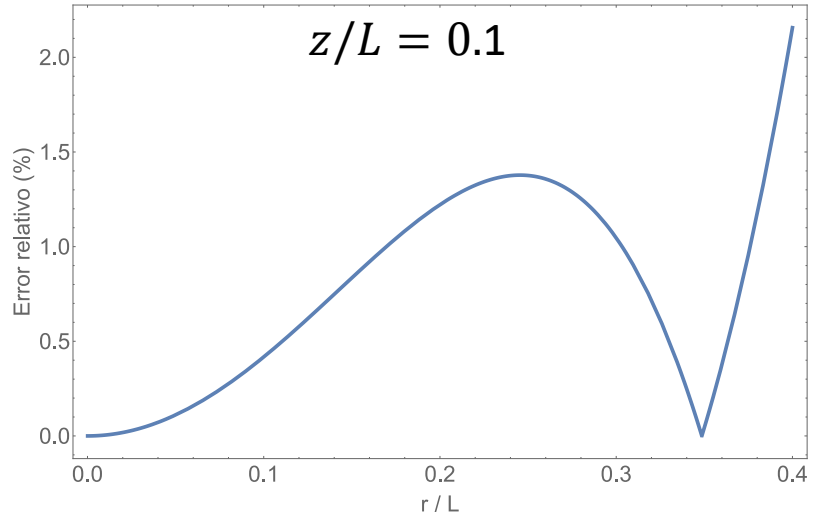

$z/L = 0.2$
E (un. arb.)
60
50
40
30
20
10
0.0
0.1
0.2
0.3
0.4
r / L

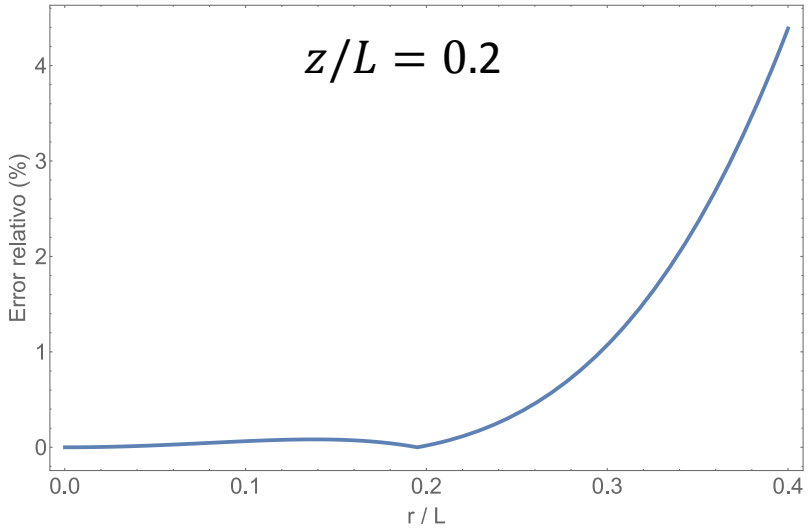

$z/L = 0.3$
E (un. arb.)
60
50
40
30
20
10
0.0
0.1
0.2
0.3
0.4
r / L

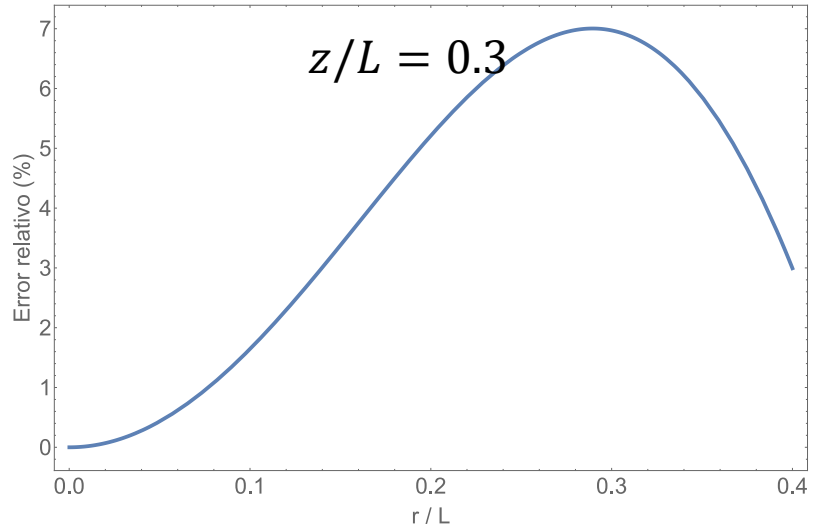
Problema 1.3

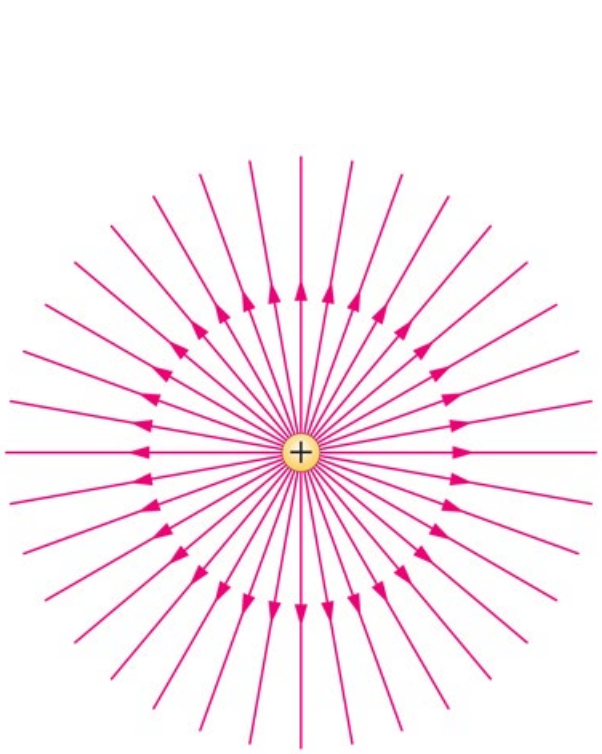

z / L
r / L
$z/L = 0.3$
$z/L = 0.2$
$z/L = 0.1$
$z/L = 0.0$

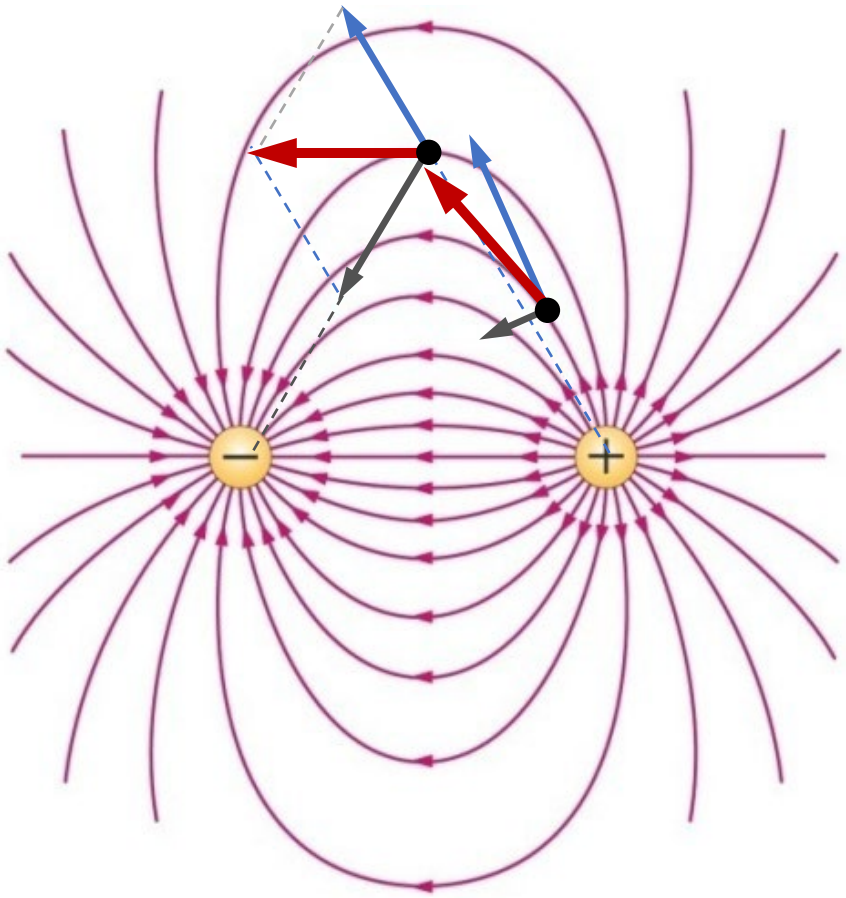

$z/L = 0.0$
Error relativo (%)
r / L

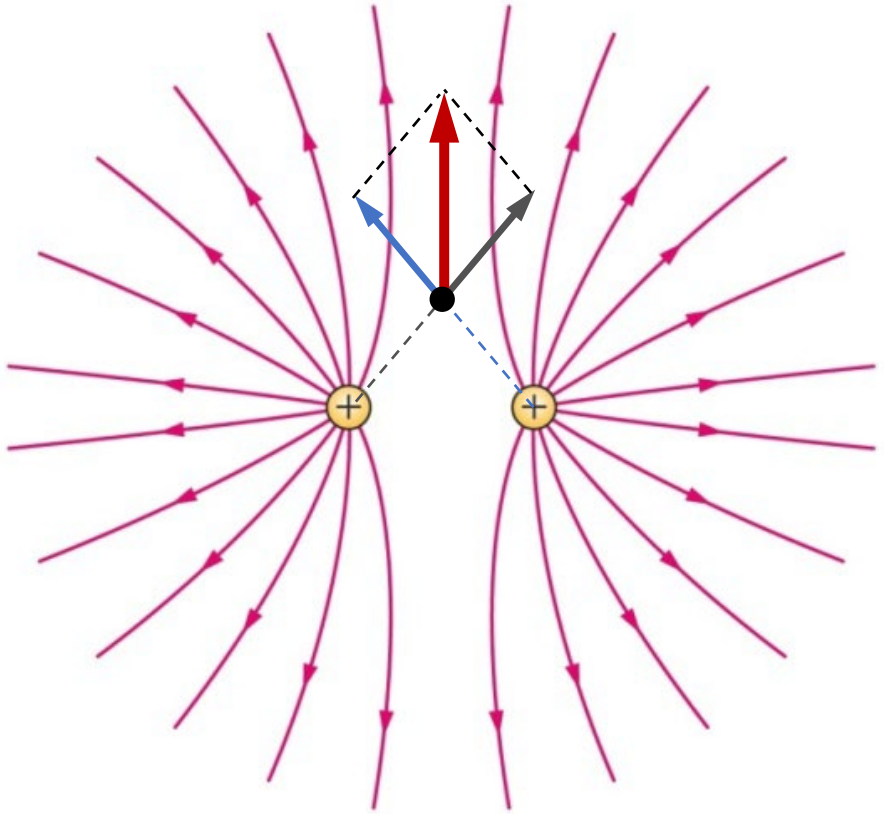

$z/L = 0.1$
Error relativo (%)
r / L

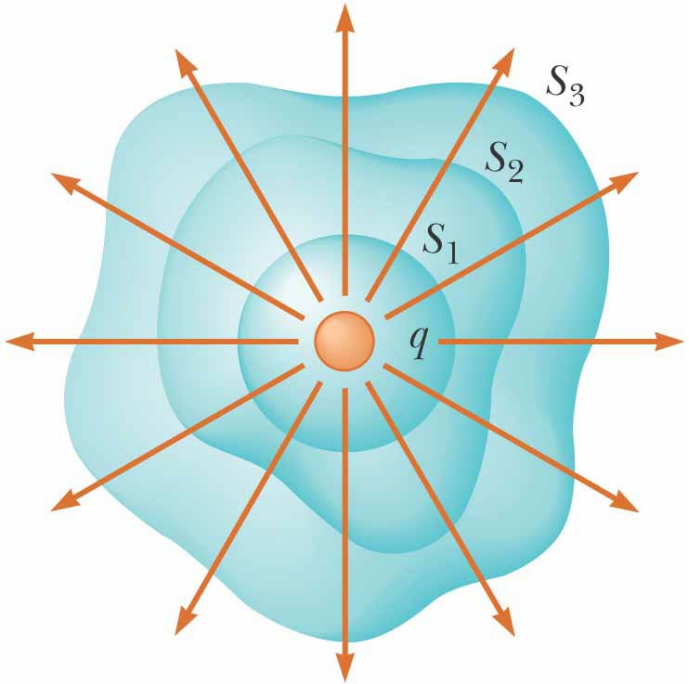

$z/L = 0.2$
Error relativo (%)
r / L

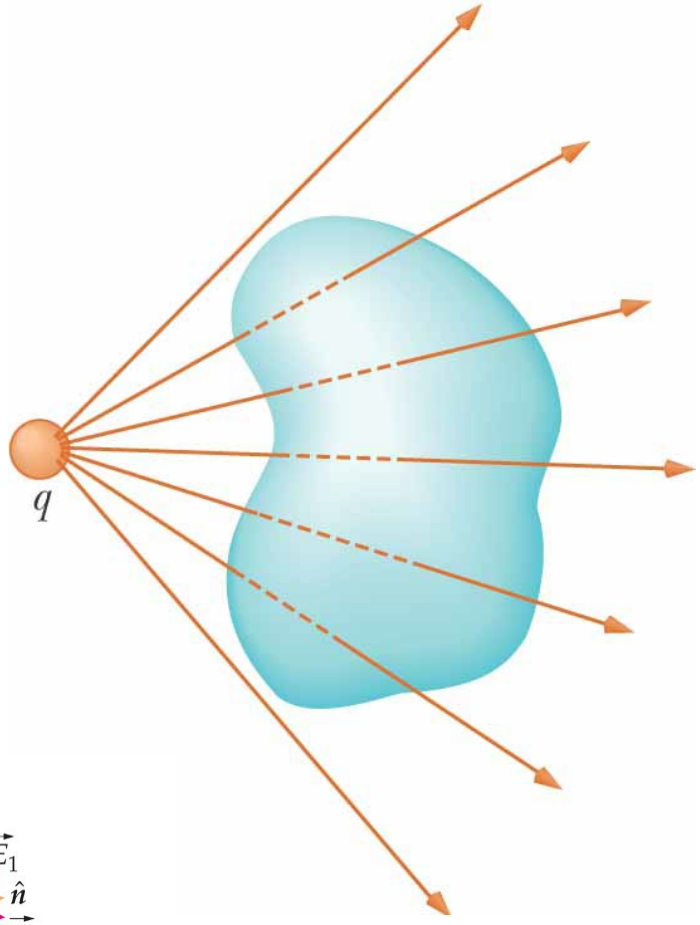

$z/L = 0.3$
Error relativo (%)
r / L

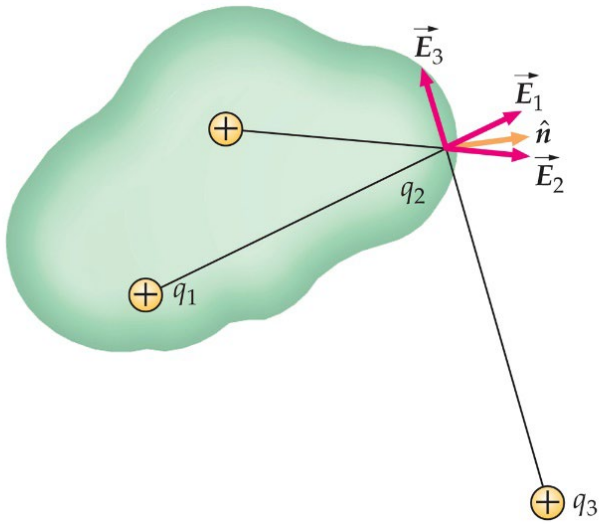


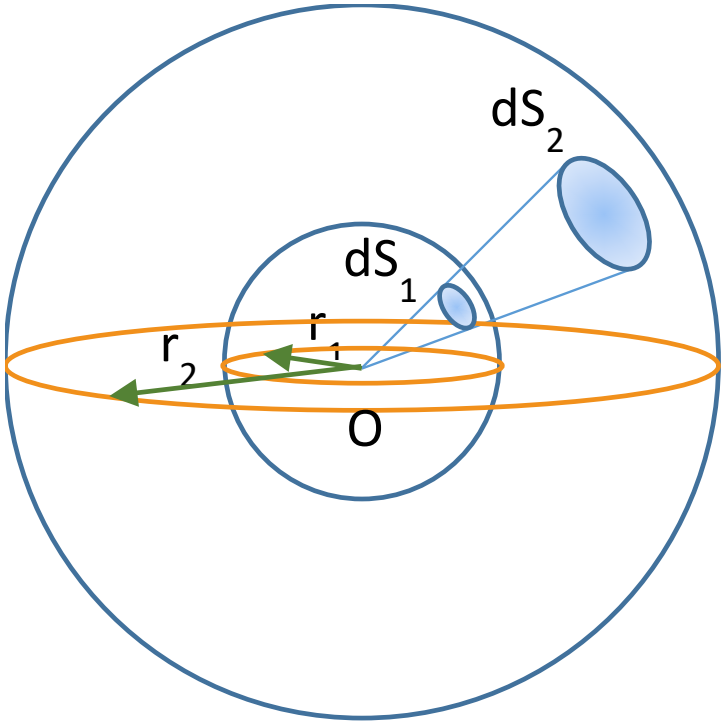


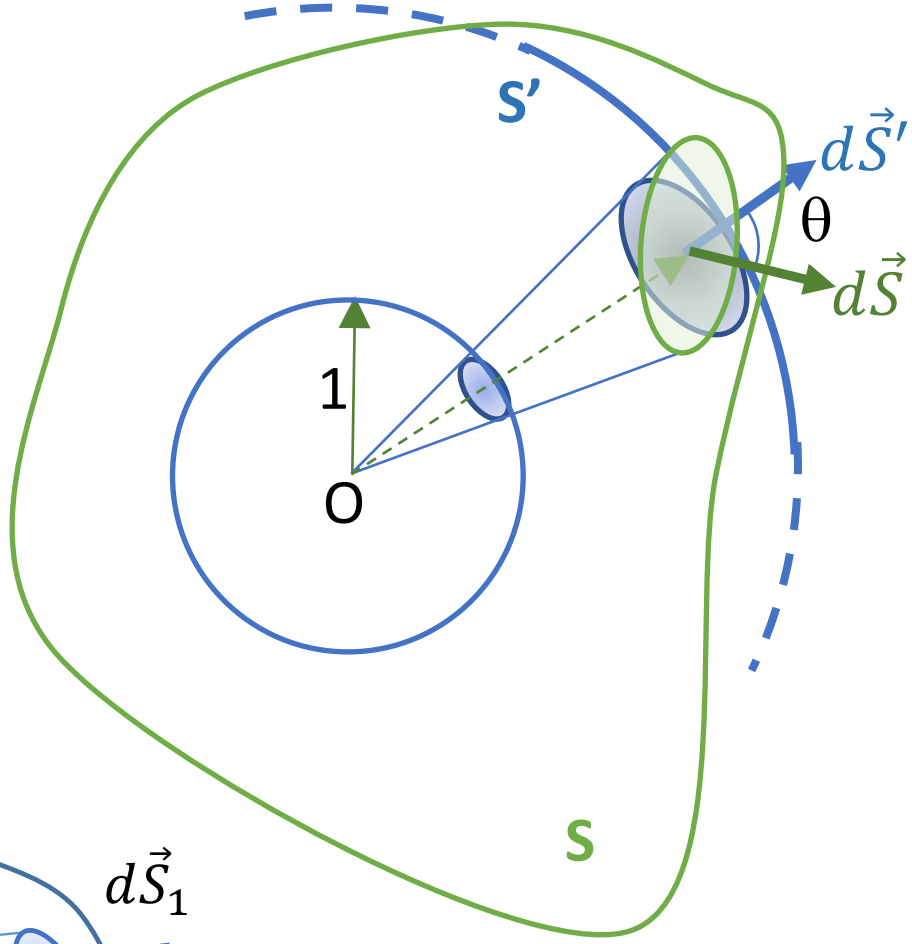


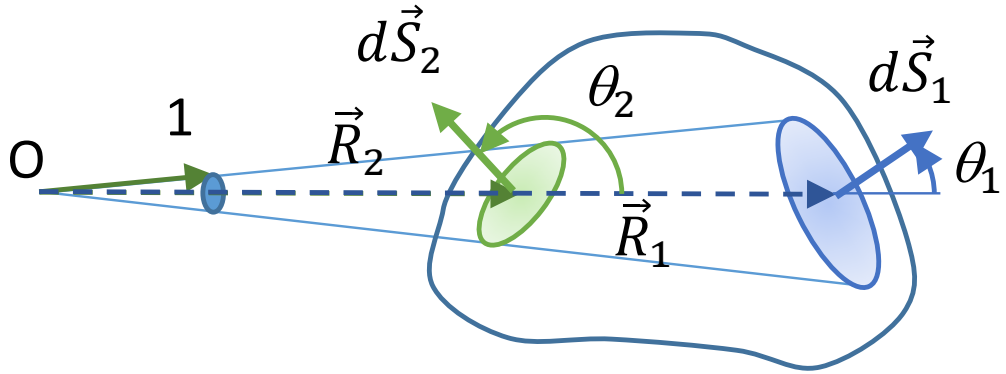

S_3
S_2
S_1
q


q

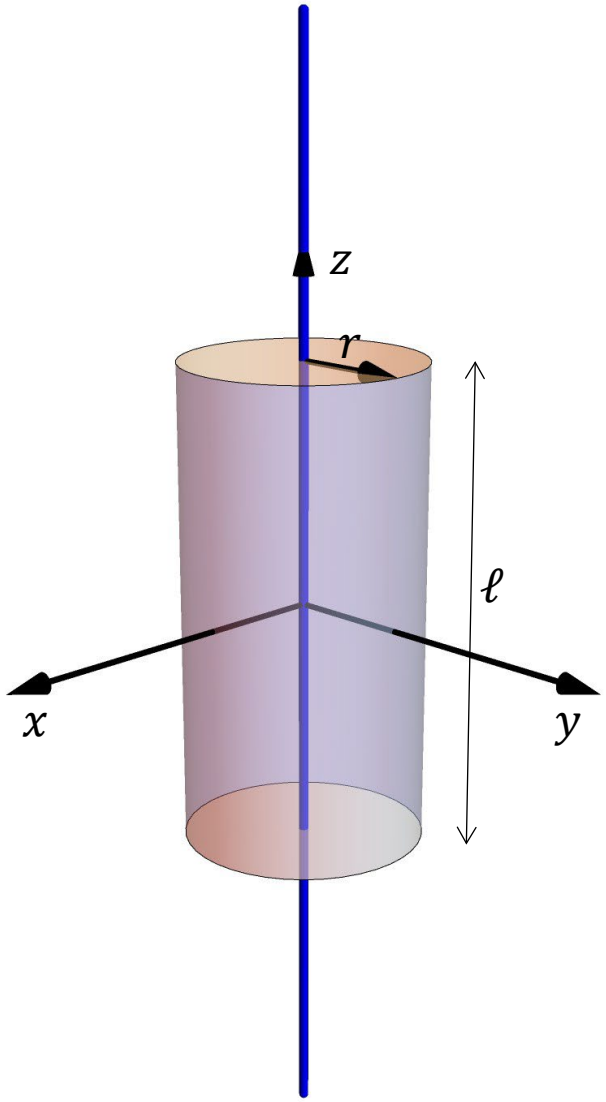

$\vec{E}_3$
$\vec{E}_1$
$\hat{n}$
$\vec{E}_2$
q_2
q_1
q_3


dS2
dS1
r2
r1
O

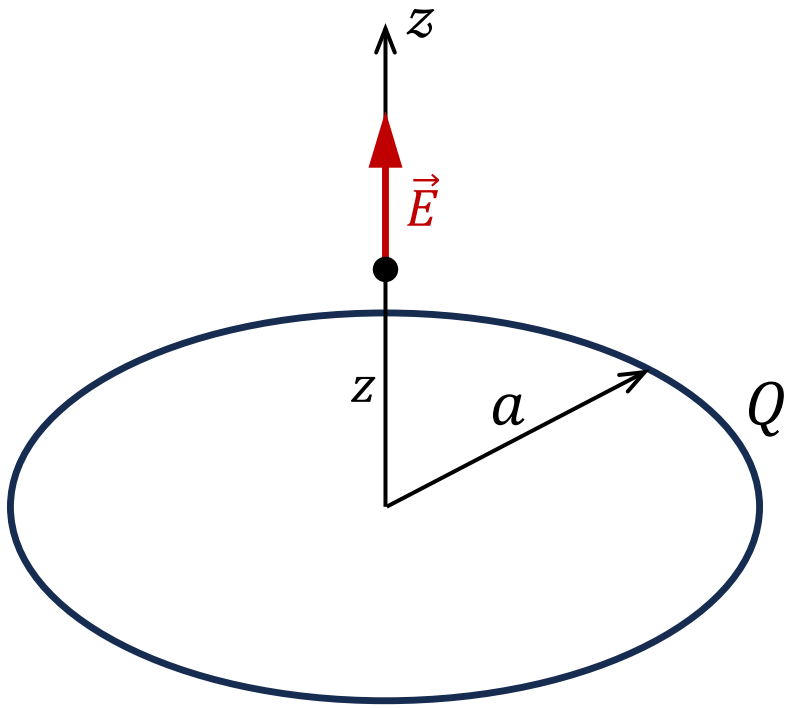

S'
dS'
θ
dS
1
O
S

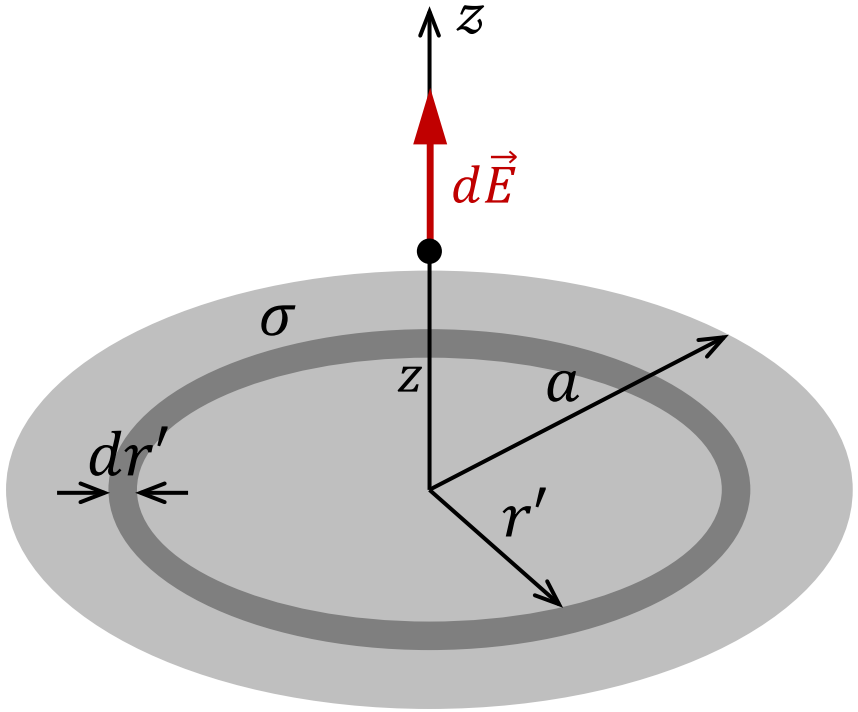

dS2
θ2
dS1
O
1
R2
θ1
R1

Problema 1.4



Problema 1.7

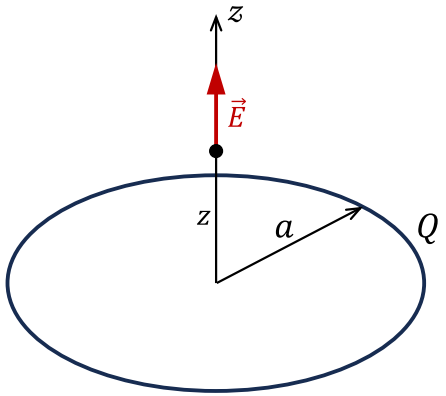


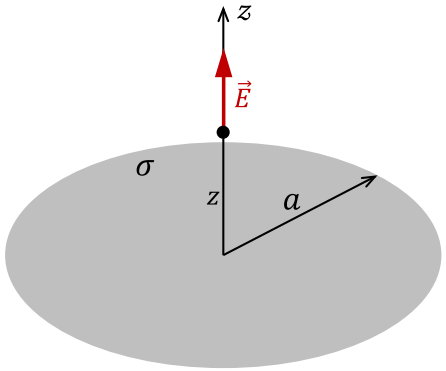


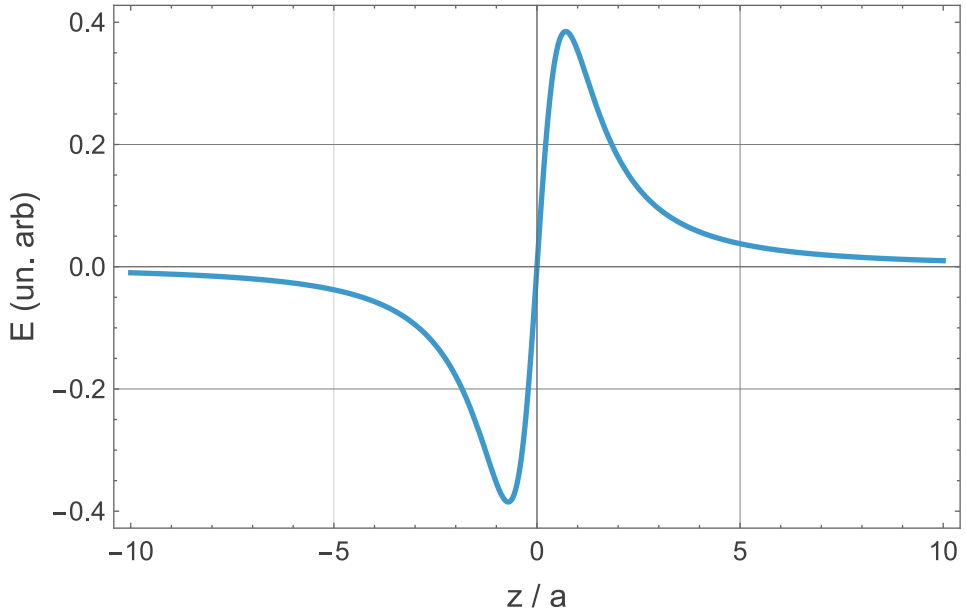
$$\vec{E}_{\text{anillo}}(z) = \frac{Qz}{4\pi\varepsilon_0(a^2+z^2)^{3/2}}\hat{u}_z$$

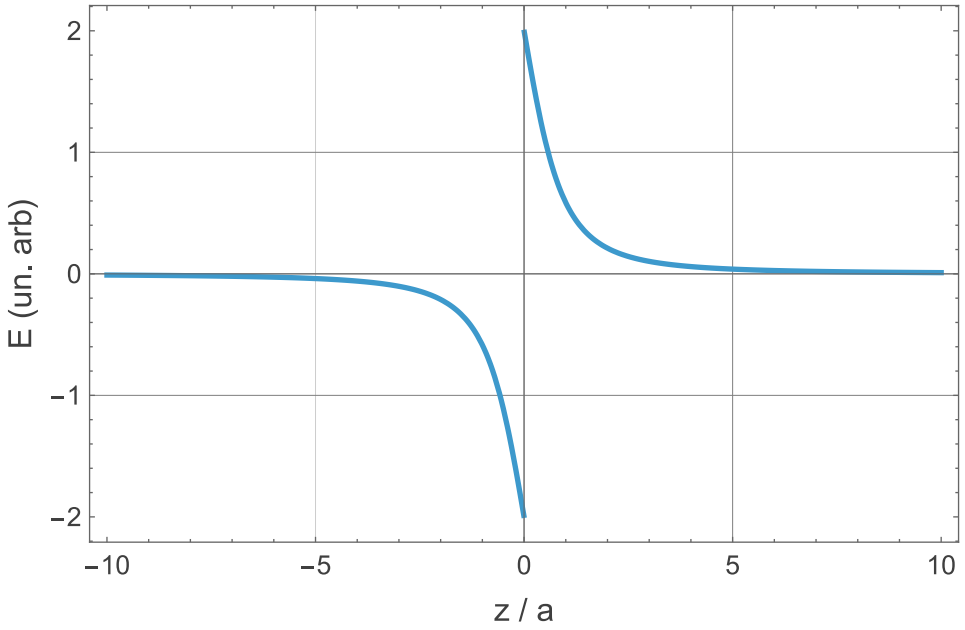
$$d\vec{E}(\text{z}) = \frac{dq\ z}{4\pi\varepsilon_0(r'^2+z^2)^{3/2}}\hat{u}_z$$

Problema 1.7





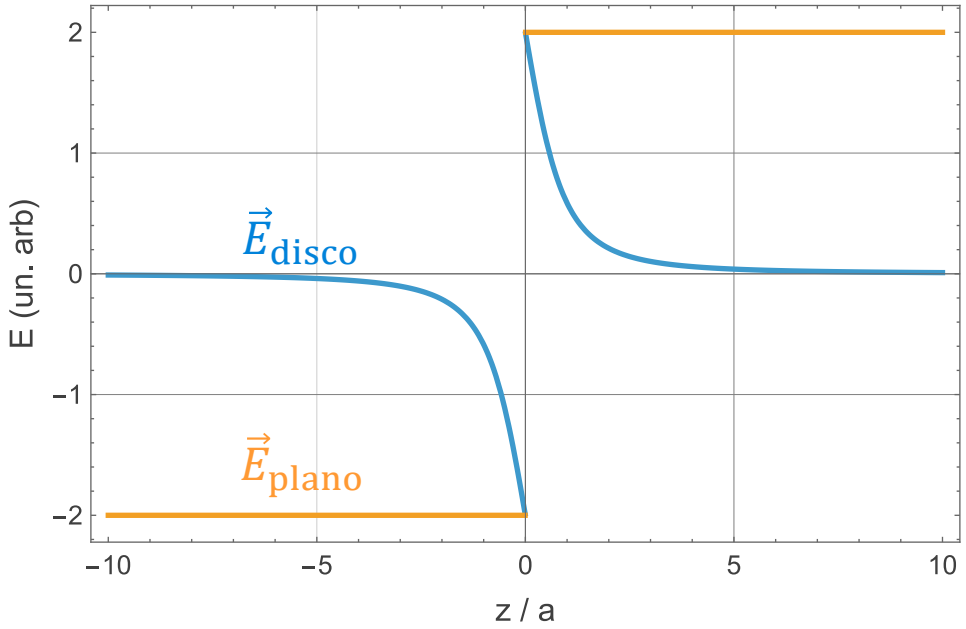


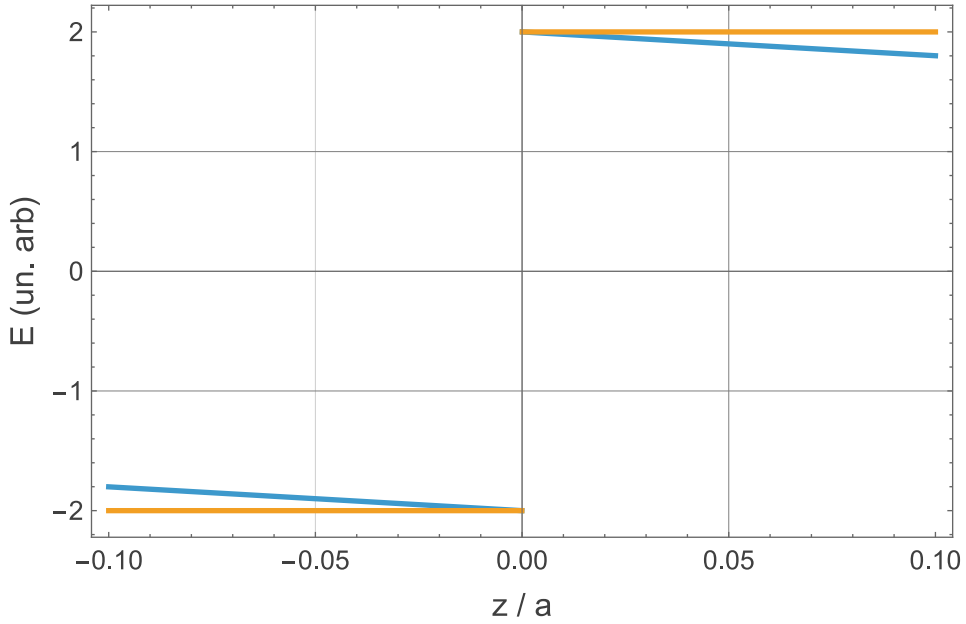


$$\vec{E}_{\text{anillo}}(\vec{r}) = \frac{Qz}{4\pi\varepsilon_0(a^2+z^2)^{3/2}}\hat{u}_z$$

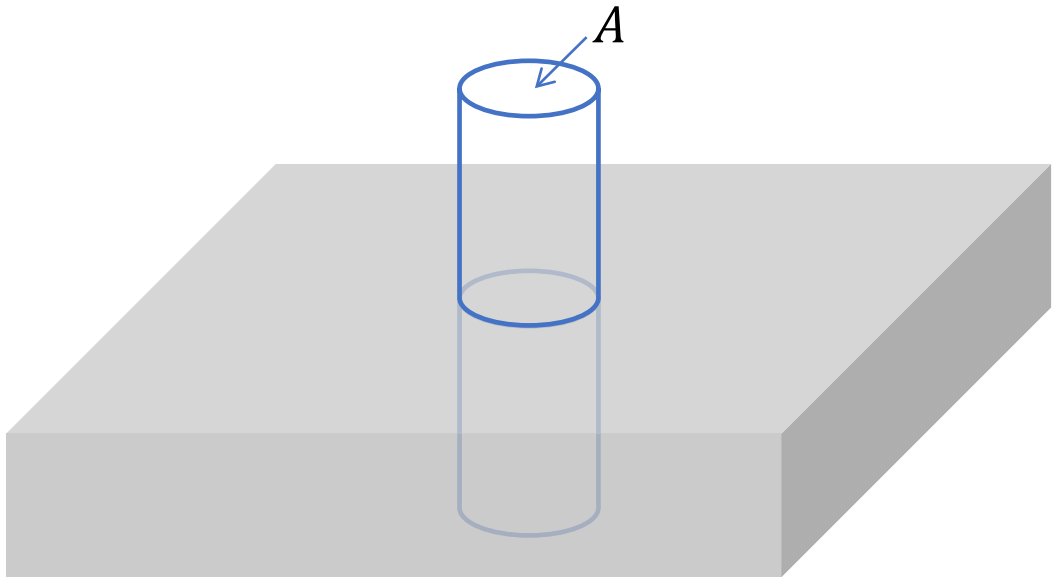
$$\vec{E}_{\text{disco}}(\vec{r}) = \frac{\sigma}{2\varepsilon_0}\left(\frac{z}{|z|} - \frac{z}{(a^2+z^2)^{1/2}}\right)\hat{u}_z$$

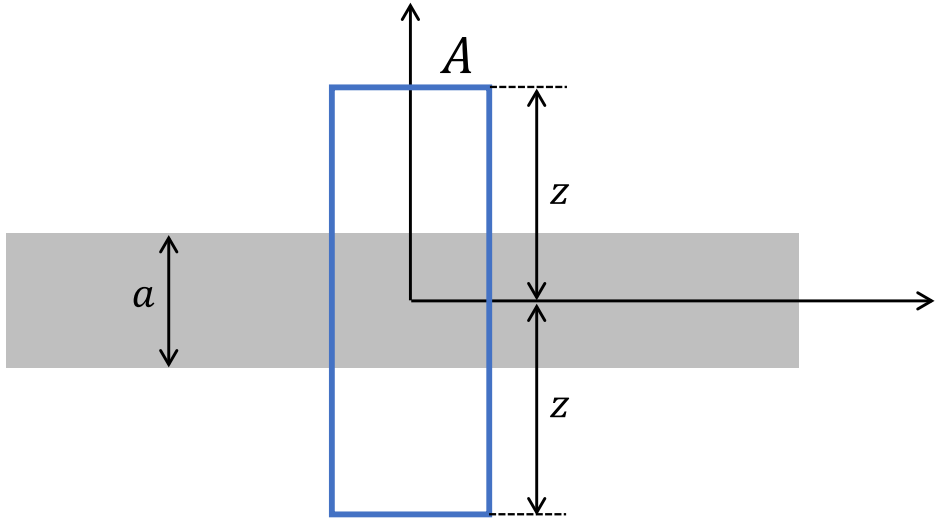
Problema 1.7


2
1
0
−1
−2
E (un. arb)
$\vec{E}_{\text{disco}}$
$\vec{E}_{\text{plano}}$
−10
−5
0
5
10
z / a

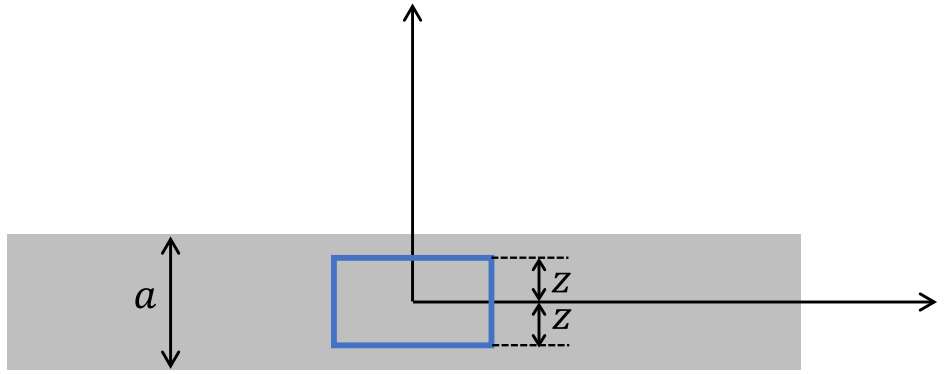

2
1
0
−1
−2
E (un. arb)
−0.10
−0.05
0.00
0.05
0.10
z / a

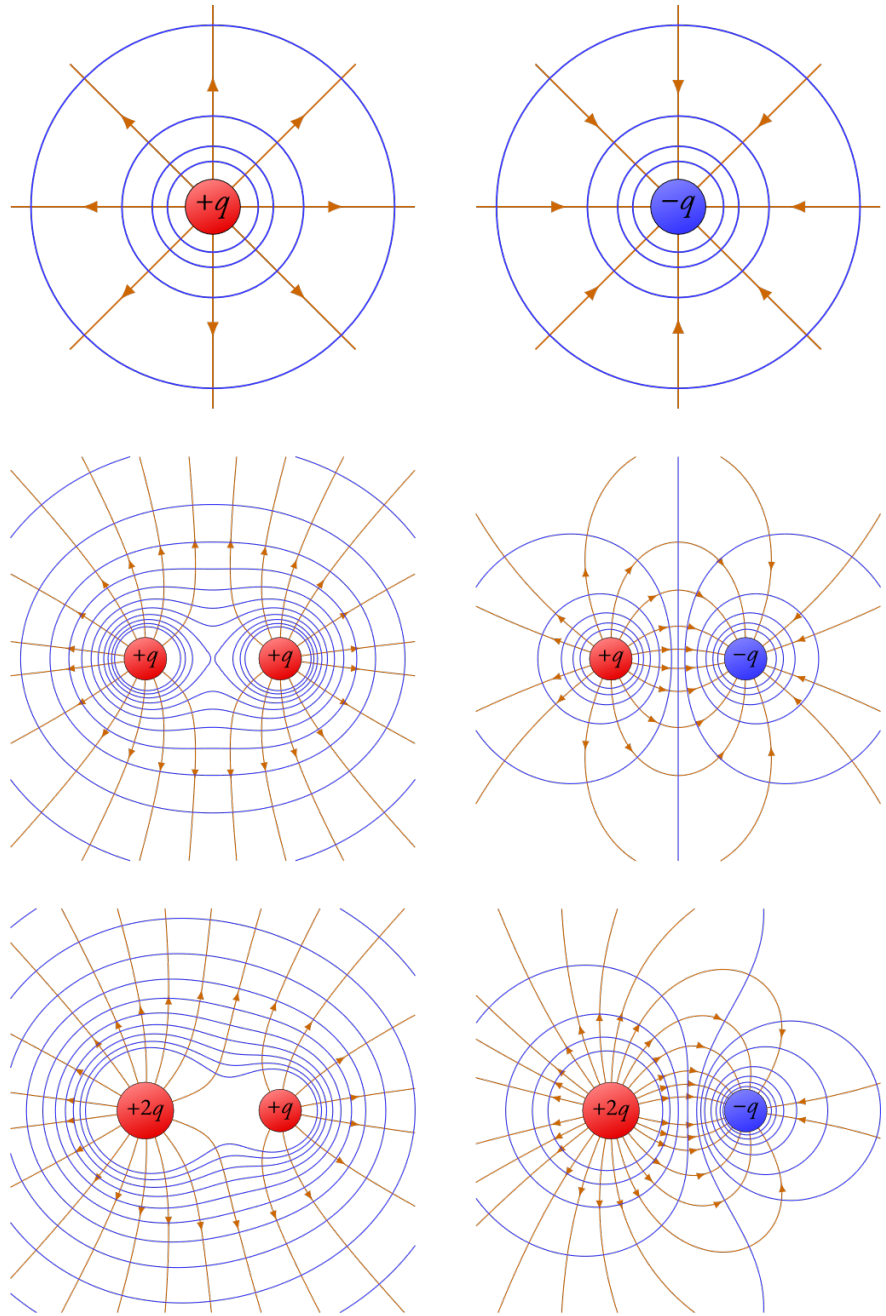
Problema 1.8



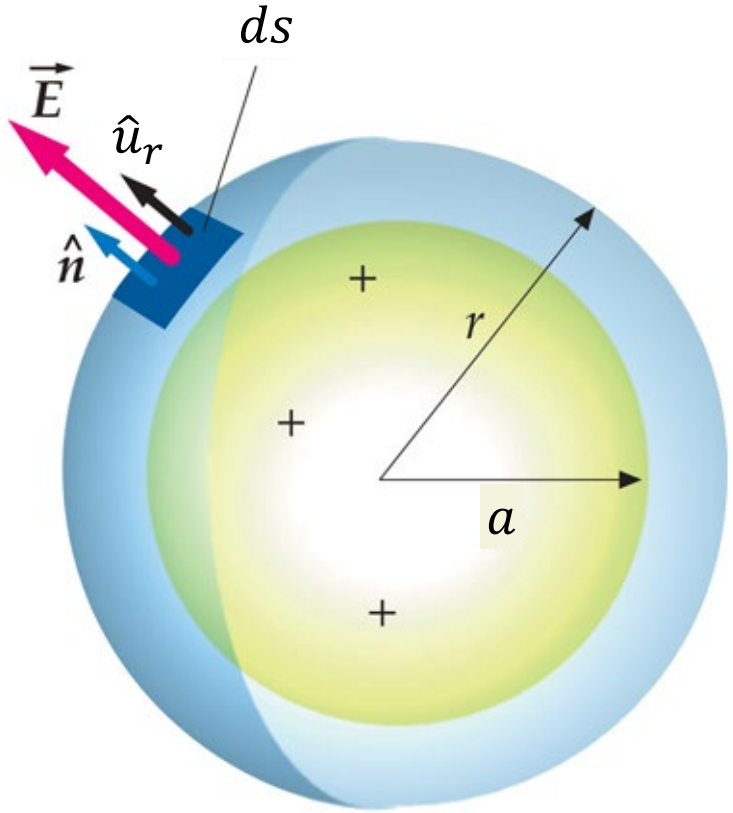


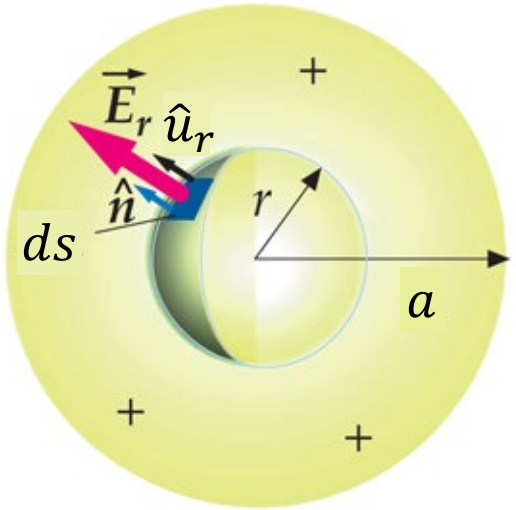




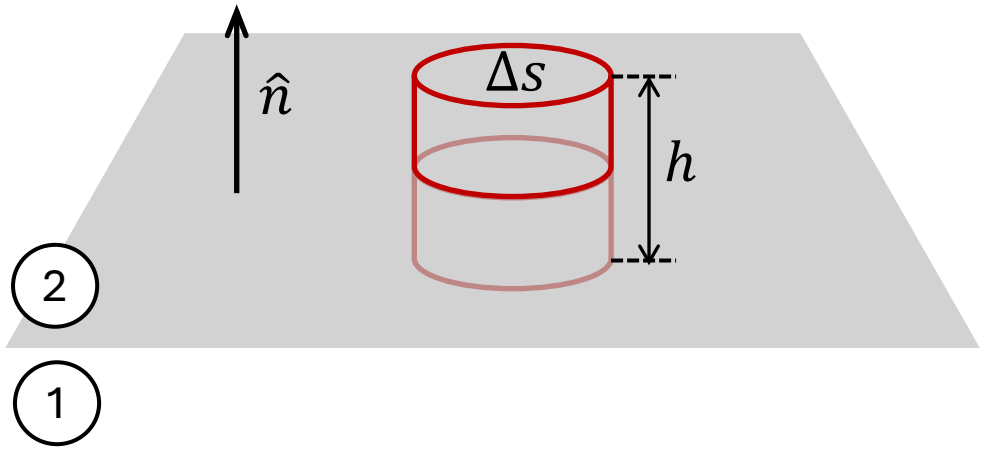

+q
−q
+q
+q
+q
−q
+2q
+q
+2q
−q

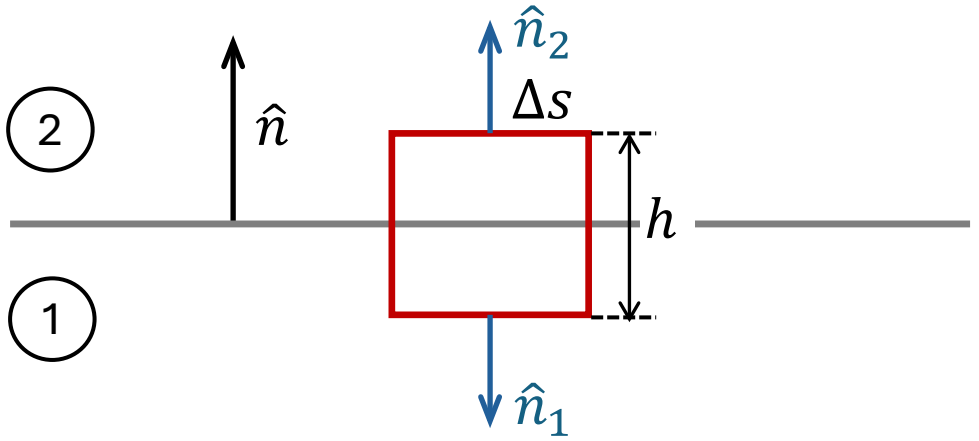
Problema 1.2





1.6. Condiciones de frontera: componente normal

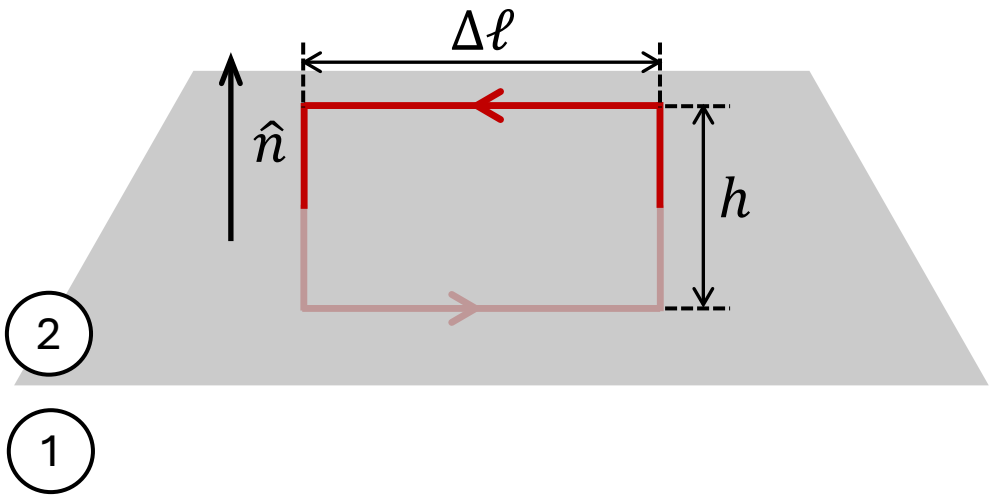


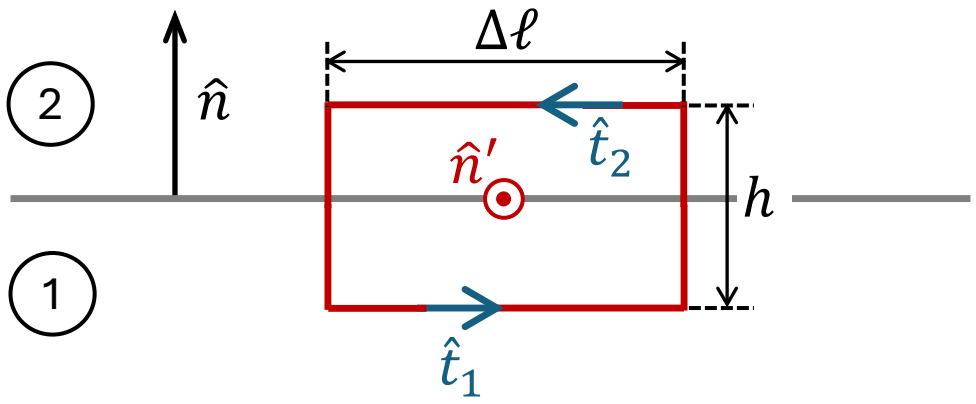


$$F_{2n} - F_{1n} = \lim_{h \to 0} (h \, \nabla \cdot \vec{F})$$

$$E_{2n} - E_{1n} = \frac{\sigma}{\varepsilon_0}$$

1.6. Condiciones de frontera: componente tangencial

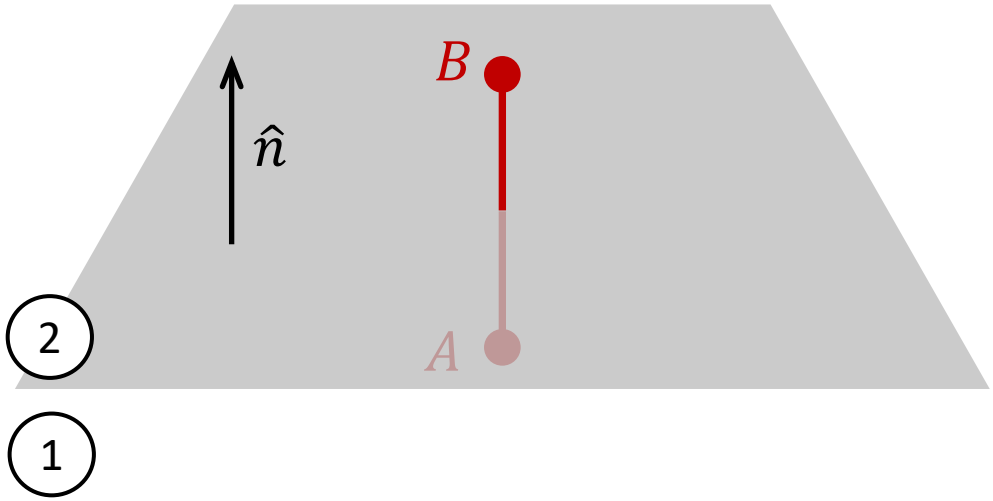


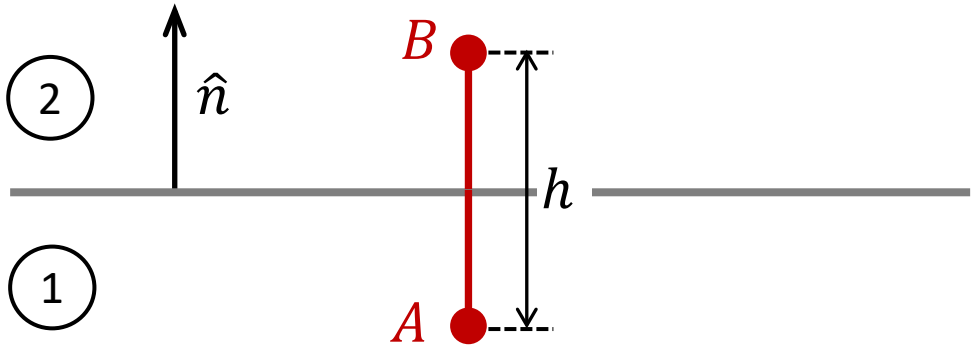


$$\vec{F}_{2t} - \vec{F}_{1t} = \lim_{h\to 0}[h(\nabla \times \vec{F}) \times \hat{n}]$$

$$\vec{E}_{2t} = \vec{E}_{1t}$$

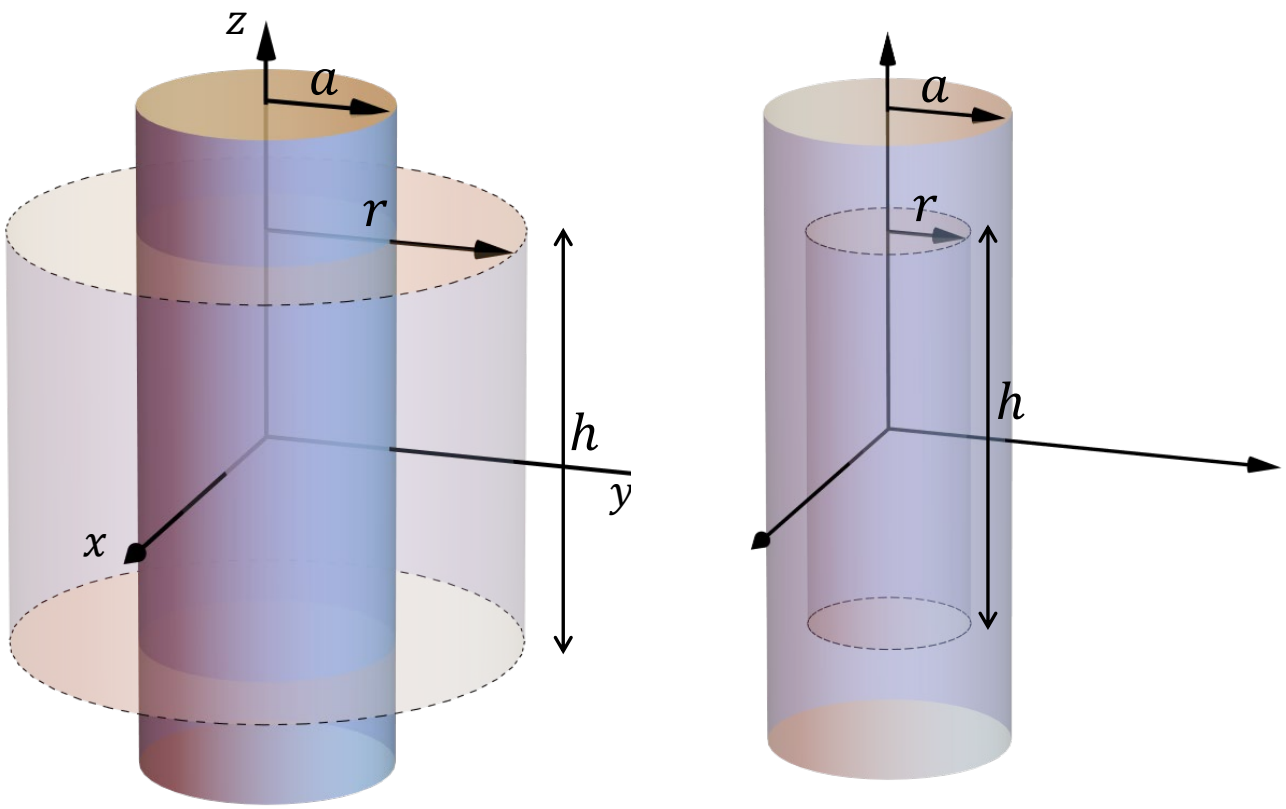
1.6. Condiciones de frontera: potencial

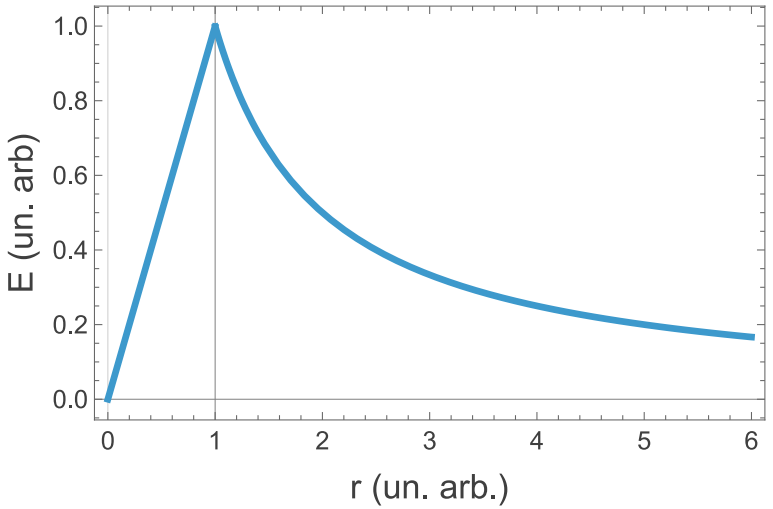


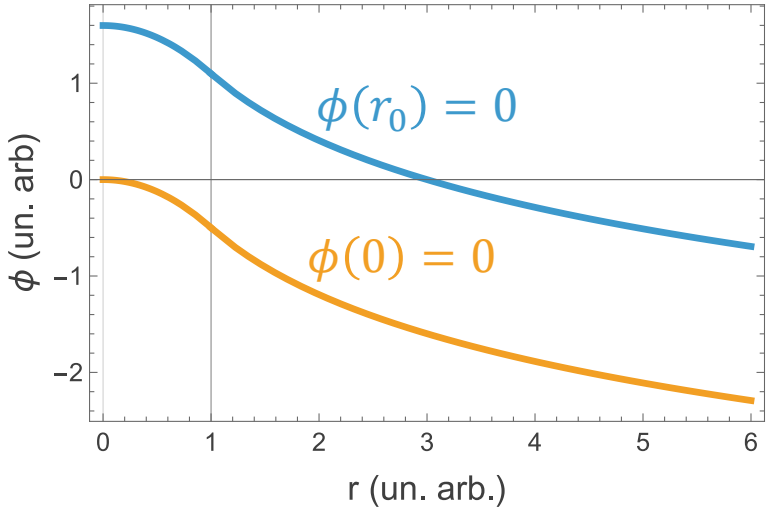


$$\phi_2 = \phi_1$$

Problema 1.10

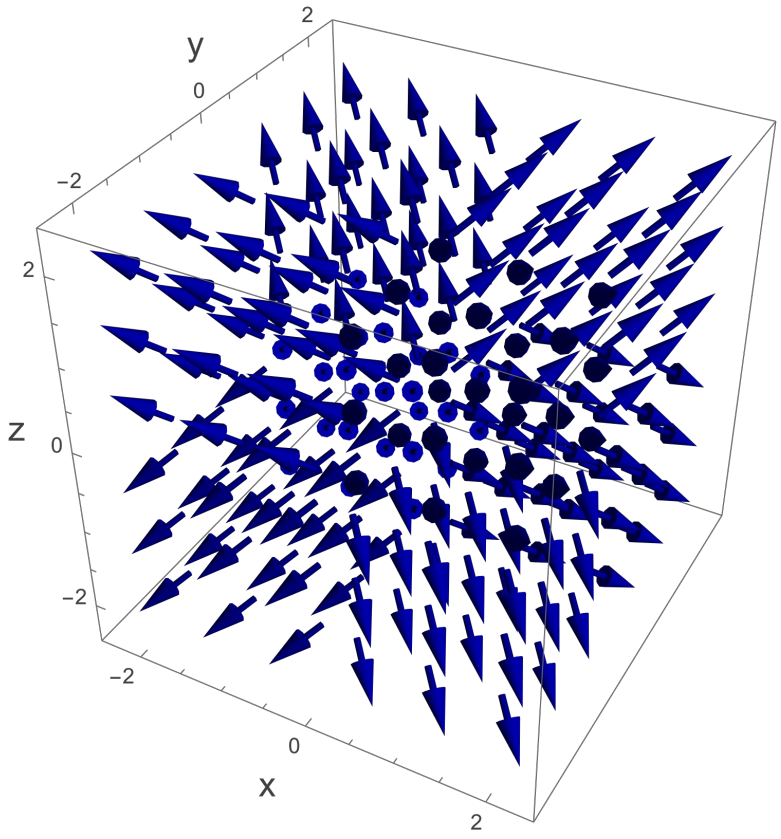


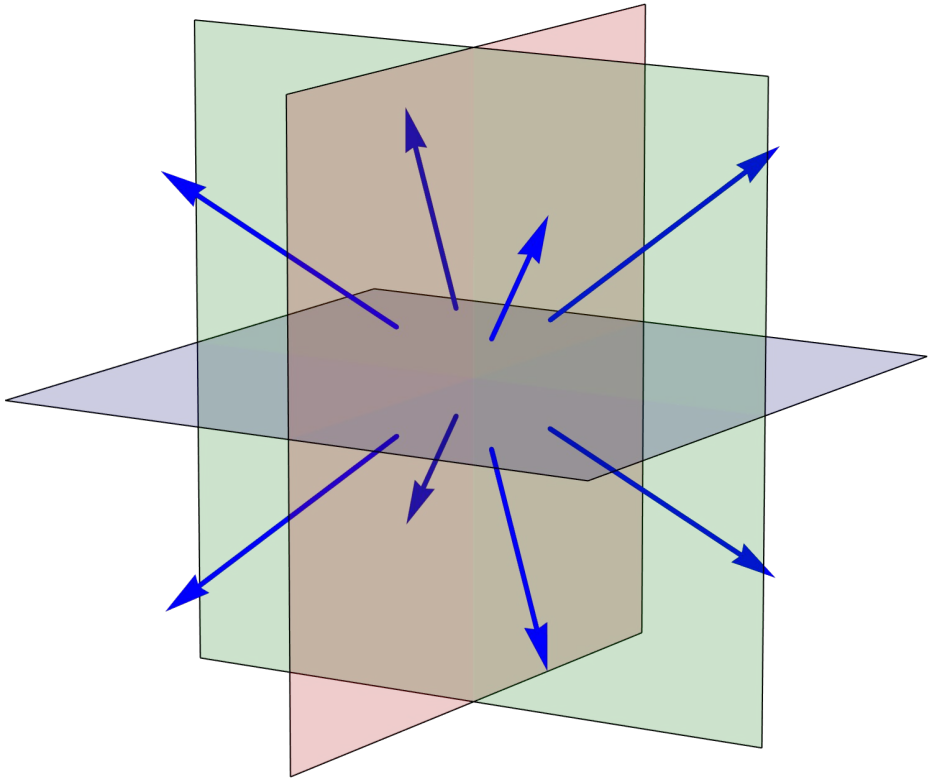




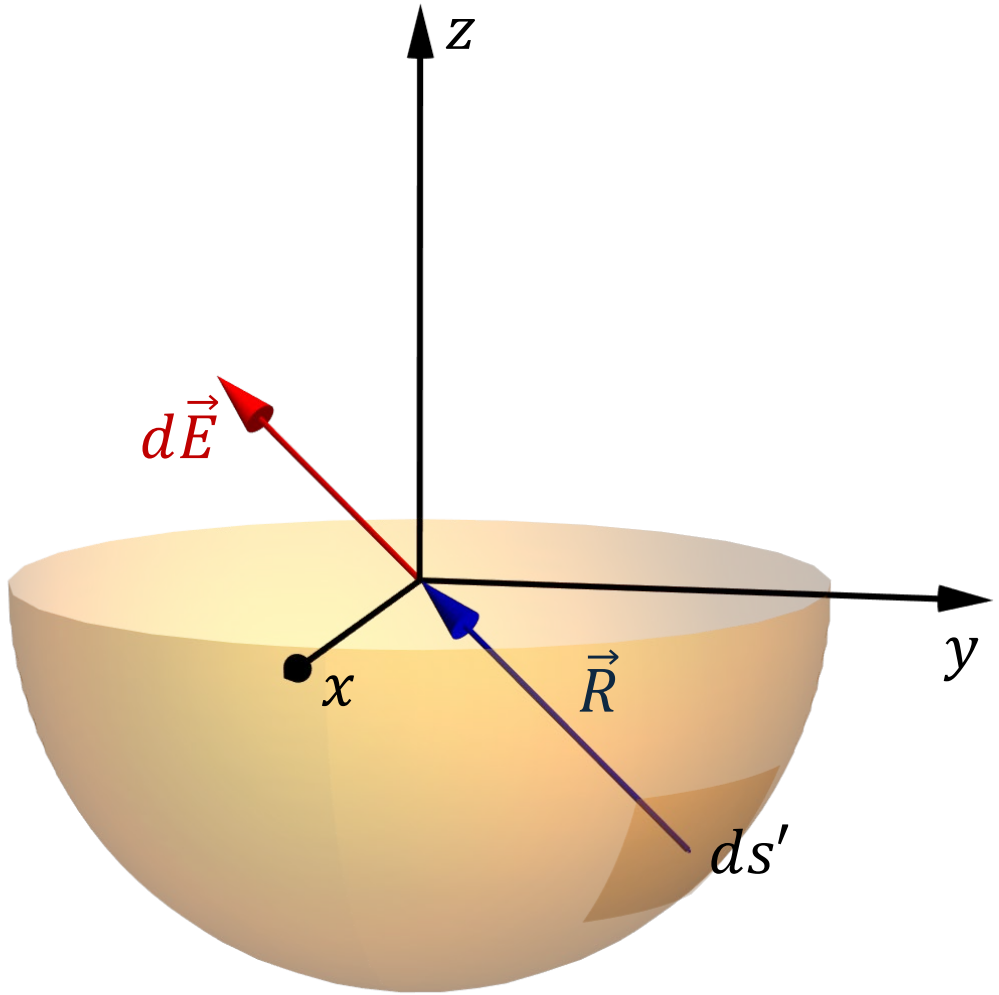
$$\rho/_{2\varepsilon_0} = 1, a = 1, r_0 = 3$$

Problema 1.11





Problema 1.12



Problema 1.1

